Head unit man-in-the-middle mode swap guide

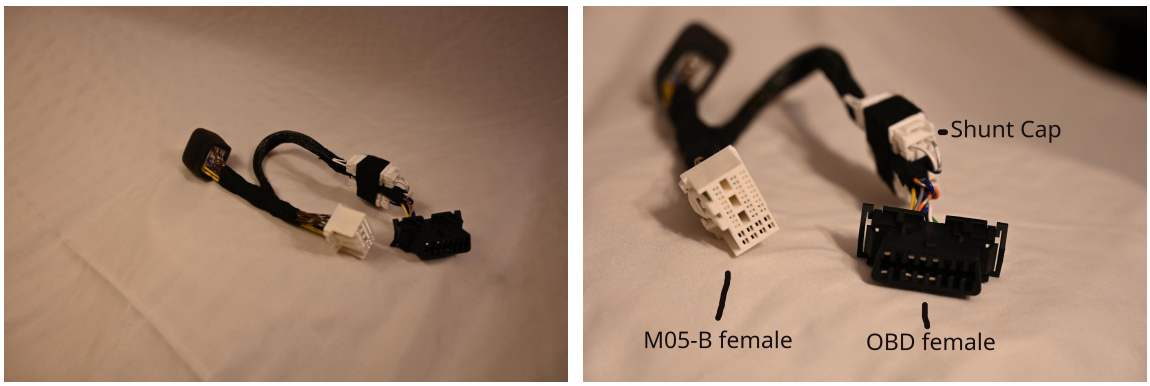
Electroniq Buttons Boutique*
(Dated: November 2, 2026)

This guide walks you through the steps needed to change the mode of your head unit man-in-the-middle (MITM) harness between MITM and parallel modes. This is a troubleshooting or advanced DIY step. See also the video guide (https://youtu.be/sWzxWfUpCfs).

I. INTRODUCTION

The head unit man-in-the-middle (MITM) harness, shown in Figure 1a, has two possible modes:

1. man-in-the-middle, or serial
2. parallel



(a) (b)

Figure 1. (a) Man-in-the-middle harness. (b) Overview of connectors showing shunt cap.

In man-in-the-middle mode, the CAN communication lines for the head unit and car are physically disconnected, and a microcontroller plugged into the OBD port is used to selectively pass messages between the two. In parallel mode, the CAN communication lines are connected normally and both CAN pin sets on the OBD port are identical and connected to both the car and head unit. The microcontroller sits on the CAN bus in the same way that a lot of ECUs are connected: in parallel, able to send and receive messages that are shared with all ECUs on the bus.

The harness is designed to be used primarily in MITM mode, but parallel mode is provided for several reasons:

1. troubleshooting
2. compatibility

If, for example, the MITM ever fails, you can easily switch to parallel mode and continue to use your head unit normally without de-installing the harness. In addition, some microcontrollers only have one CAN pin set and cannot do forwarding. Parallel mode is provided for compatibility in these cases.

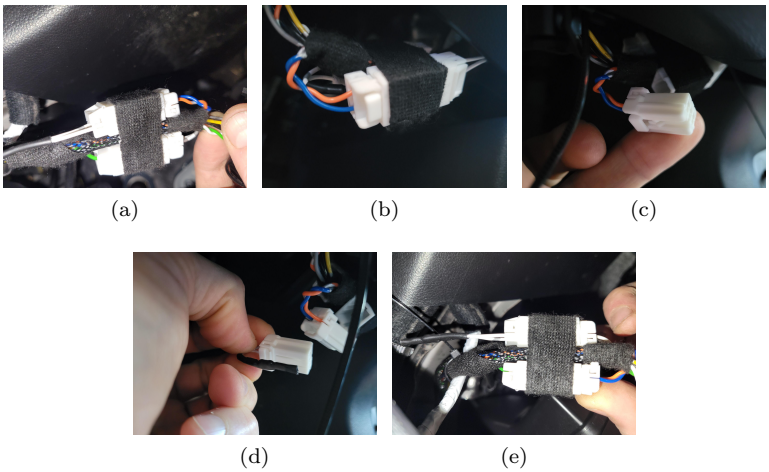
II. HOW TO SWITCH

To switch, you just have to swap the female shunt and shunt caps, which sit closest to the OBD port. To move from MITM to parallel, do the following:

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1. Check that current harness mode is MITM (Figure 2a)
2. Disconnect the female shunt (orange and blue wires, labeled "B1") (Figure 2c)
3. Disconnect the female shunt cap (white wires, labeled "A2") (Figure 2d)
4. Connect the female shunt into the male shunt (green and white wires, labeled "A1") (Figure 2e)
5. Connect the female shunt cap into the male shunt cap (white wires, labeled "B2") (Figure 2e)

To reverse the mode, just reverse these instructions!



(a) (b) (c)

(d) (e)

Figure 2. (a-b) Female shunt cap: initial MITM mode. (c) Disconnecting female shunt. (d) Disconnecting female shunt cap. (e) Parallel mode.